

ALLEN TU

Ph.D. Student, Computer Science — University of Maryland

Research Areas: Reliable and Scalable Vision Systems, 3D/4D Reconstruction, Biometrics, Generative Models

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SELECTED PUBLICATIONS

1. H. Ying, **Allen Tu**, J. Wu, T. Goldstein, and M. Zwicker, ‘UniBRep: Learning Unified Geometry and Topology for Image-conditioned B-Rep Generation’. *Under review*, 2026.
2. **Allen Tu**^{*}, H. Ying^{*}, A. Hanson, Y. Lee, T. Goldstein, and M. Zwicker, ‘[Speede3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping](#)’. *CVPR*, 2026.
3. P. Asthana, A. Hanson, **Allen Tu**, T. Goldstein, M. Zwicker, and A. Varshney, ‘[SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting](#)’. *CVPR*, 2026.
4. **Allen Tu**, K. Narayan, J. Gleason, J. Xu, M. Meyn, and V. Patel, ‘[TransFIRA: Transfer Learning for Face Image Recognizability Assessment](#)’. *FG*, 2026 (**Oral Presentation**).
5. A. Hanson^{*}, **Allen Tu**^{*}, V. Singla, M. Jayawardhana, M. Zwicker, and T. Goldstein, ‘[PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting](#)’. *CVPR*, 2025.
6. A. Hanson, **Allen Tu**, G. Lin, V. Singla, M. Zwicker, and T. Goldstein, ‘[Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives](#)’. *CVPR*, 2025.

^{*} denotes equal contribution.

EDUCATION

University of Maryland, College Park

Ph.D. Student, Computer Science

College Park, MD

January 2025 – May 2028

- **Advisor:** Professor Tom Goldstein

- **Collaborators:** Professors Matthias Zwicker, Vishal M. Patel, Rama Chellappa

B.S./M.S. in Computer Science, Minor in Statistics

August 2019 – December 2024

RESEARCH EXPERIENCE

Amazon

Incoming Applied Science Intern

Bellevue, WA

June 2026 – October 2026

- Joining the Last Hundred Yards team to research 3D reconstruction for robotics

University of Maryland Institute of Advanced Computer Studies

Graduate Research Assistant

College Park, MD

August 2023 – Present

- Developing scalable 3D reconstruction pipelines deployed for unconstrained novel-view synthesis in real-world environments under the IARPA Walk-through Rendering from Images of Varying Altitude ([WRIVA](#)) program
- Created pruning, rasterization, and motion distillation methods that accelerate static and dynamic 3D Gaussian Splatting, producing representations with over 10× fewer primitives while preserving visual fidelity [2, 5, 6]
- Introduced uncertainty-aware diffusion and super-resolution priors into 3D reconstruction pipelines to improve fidelity under sparse and low-resolution supervision while mitigating artifacts and cross-view inconsistencies [3]

Systems & Technology Research

Computer Vision Research Intern

Arlington, VA

May 2025 – January 2026

- Engineered multimodal biometric systems fusing face, body, and gait recognition for robust identification under severe conditions in the IARPA Biometric Recognition and Identification at Altitude and Range ([BRIAR](#)) program
- Proposed an explainable transfer-learning method for face image quality assessment (FIQA), enabling recognizability-aware probe filtering and weighted template aggregation for face and body recognition [4]
- Demonstrated state-of-the-art template-based recognition on the proprietary BRIAR surveillance benchmark, improving TAR from 0.43 to 0.87 at 1e-3 FMR using an encoder trained solely on public data [4]

Systems & Technology Research

Arlington, VA

Computer Vision Research Intern

May 2024 – August 2024

- Achieved 0.86 correlation with ground-truth probe-to-gallery similarity and state-of-the-art ERC performance for face image quality estimation (FIQA), outperforming prior methods on image-level evaluation [4]
- Designed a cluster-and-aggregate network that filters low-importance frame embeddings and fuses the remaining informative ones into a single video template, improving face recognition accuracy by up to 12%

Computer Vision Research Co-op

January 2023 – August 2023

- Implemented a mixed-voting ensemble that improved open-set search performance by 24% for face recognition, 39% for body recognition, and 49% for fused multimodal biometric recognition
- Trained a self-supervised Barlow Twins model robust to challenges like extreme distance variation and atmospheric interference, improving face recognition accuracy by 52% in severe operating conditions

Computer Vision Research Intern

June 2022 – August 2022

- Integrated pose estimation, neural novel-view synthesis, semantic segmentation, and generative inpainting to synthesize realistic cross-subject garment transfers with strong biometric identity preservation
- Augmented real training data with garment-transfer images to overcome limited garment diversity and train clothing-invariant whole-body recognition models, improving robustness to cross-garment variation

Undergraduate Researcher

College Park, MD

University of Maryland Department of Computer Science

January 2021 – December 2022

- Analyzed 2.3 million dataset-visualization pairs to identify bias in machine learning visualization recommenders, linking model behavior to user-driven design patterns and proposing data-centric mitigation strategies [6]

RESEARCH PROJECTS

IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA)

University of Maryland Institute of Advanced Computer Studies

August 2023 – Present

Unconstrained 3D reconstruction and novel view synthesis in challenging real-world environments.

- Efficient rendering, compression, and training methods for 3D and 4D Gaussian Splatting (3DGS/4DGS) [2, 5, 6]
- Multi-view consistent super-resolution for training 3DGS with low-resolution imagery [3]
- Image, video, and multi-view diffusion model priors for sparse-view 3D reconstruction
- Uncertainty quantification algorithms for 3DGS and Neural Radiance Fields (NeRF) [5]

IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR)

Systems & Technology Research

June 2022 – January 2026

Multimodal, opportunistic fusion of incomplete face, body, and gait information in severe operational conditions.

- Transfer Learning for Face Image Recognizability Assessment [4]
- Learned Frame Feature Aggregation for Face Recognition with Low-Quality Video
- Operating Condition-Invariant Barlow Twins and Multimodal Ensembling
- Style-Based Appearance Flow for Clothing-Robust Body Representation Learning

HONORS & AWARDS

FG 2026 Doctoral Consortium: Selective Computation for Reliable and Scalable Vision Systems

University of Maryland Travel Awards: Jacob K. Goldhaber Travel Grant, CMSC PhD Travel Grant, ICSSA

SERVICE

SPAR-3D: Security, Privacy, and Adversarial Robustness in 3D Generative Vision Models

Workshop Co-organizer

CVPR 2026

- Co-organized the first CVPR workshop on security, privacy, and robustness in 3D generative vision models, coordinating peer review, securing \$1K in industry sponsorship, and moderating keynote and Q&A sessions

Reviewer: SIGGRAPH Asia, IEEE TBIOM, CVPR Workshops, FG, IEEE TCSVT

TEACHING EXPERIENCE

Peer Research Mentor

College Park, MD

FIRE: Capital One Machine Learning

January 2021 – December 2022

- Mentored 80 undergraduate students across two cohorts in a selective multi-semester machine learning research program, guiding students from foundational training to independent research project execution
- Designed and delivered a project-driven curriculum spanning data processing, model development, training, evaluation, and deployment using modern deep learning frameworks and recent research advances
- Advised student teams end-to-end on collaborative research projects, leading to presentations in areas including text-to-audio generation, image super-resolution, and face attribute recognition

TECHNICAL SKILLS

Languages: Python, CUDA, C/C++, C#, Java, SQL, Shell

Frameworks & Tools: PyTorch, TensorFlow, NumPy, SciPy, OpenCV, scikit-learn

Research Areas: 3D/4D Reconstruction, 3D Gaussian Splatting, Neural Radiance Fields (NeRF), Differentiable Rendering, Biometrics, Diffusion Models, Vision Transformers (ViT/Swin), Uncertainty Quantification, Scalable Vision Systems, GPU Programming